

Synthesis of Unsymmetrical Arylheteroarylmethanes by Direct “On Water” Cross-Coupling between Benzylic and Heteroaromatic Halides

Valeria Krasovskaya, Arkady Krasovskiy, and Bruce H. Lipshutz*^[a]

Dedicated to Professor Eun Lee on the occasion of his retirement and 65th birthday

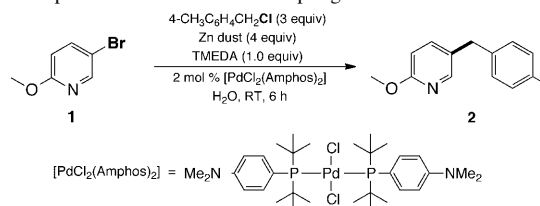
Unsymmetrical arylheteroarylmethanes have been shown to possess important biological and medicinal properties.^[1] These compounds also frequently appear as subunits in various supramolecular structures.^[2] One method for synthesizing arylheteroarylmethanes is the Friedel–Crafts alkylation of aromatic compounds with benzylic halides.^[3] Alternatively, transition-metal-catalyzed cross-coupling reactions^[4] have been employed between heteroaryl halides and benzylic metals.^[5] Analogous reactions of heteroaryl metals with benzyl halides/phosphates/carbonates are less-frequently seen.^[6] Nonetheless, in both cases, organometallic reagents are necessary, and are prepared in separate steps. Avoiding the preliminary formation of reactive organometallic species not only saves one step, but also eliminates potential issues associated with storage and handling. This advantage could be important, especially for flammable Grignard reagents (Kumada cross-coupling), and moisture-sensitive organozinc reagents (Negishi cross-coupling). Consequently, there is interest in developing cross-coupling reactions without the pre-formation of organometallic reagents. During the course of our work on direct sp^3 – sp^2 cross-coupling reactions,^[7] we reported a convenient approach for the synthesis of diarylmethanes.^[8] Herein, we describe related couplings between benzylic halides and heteroaromatic halides performed “on water”^[9] at room temperature, leading to valuable unsymmetrical arylheteroarylmethanes in good to excellent yields.

Our initial attempts focused on optimization of the reaction conditions for the coupling of representative 1-(chloromethyl)-4-methylbenzene with 2-methoxy-5-bromopyridine (**1**) to afford unsymmetrical arylheteroarylmethane **2** using

$[PdCl_2(Amphos)_2]$ ^[10] as the catalyst (Table 1). We found that **1** efficiently afforded the coupled product within six hours, on water, at room temperature, using 2 mol % $[PdCl_2(Amphos)_2]$, zinc dust (4 equiv), and *N,N,N',N'*-tetramethylethylenediamine (TMEDA; 1 equiv).^[7,8]

Three equivalents of 1-(chloromethyl)-4-methylbenzene were required to achieve full conversion along with a good yield of isolated product (Table 1, entry 1). Using two equivalents, competitive insertion of zinc metal into pyridine **1** occurred, leading to protio-quenched product 2-methoxypyridine (Table 1, entry 2). Unlike the case of cross-couplings of aryl bromides,^[8] one equivalent of TMEDA and 2 mol % of the palladium catalyst were needed to reach full conversion (Table 1, entries 3–5). Although the reaction was faster in the presence of surfactant (complete within four hours; Table 1, entries 7 and 8),^[7,11] pure water alone was used for reasons of simplicity.

Table 1. Optimization of the cross-coupling with bromide **1**.^[a]



Entry	Modification of original conditions	Conversion [%] ^[b]
1	none	100 (88) ^[d]
2	2 equiv BnCl	100 ^[c] (79) ^[d]
3	TMEDA (0.50)	90
4	TMEDA (0.25)	81
5	1 mol % $[PdCl_2(Amphos)_2]$	92
6	Zn powder	96 (83) ^[d]
7	2 wt % PTS/H ₂ O, 4 h	100
8	2 wt % TPGS-750-M/H ₂ O, 4 h	100

[a] Conditions : 1-(chloromethyl)-4-methylbenzene (3 mmol), 2-methoxy-5-bromo-pyridine (1 mmol), zinc dust (4 mmol), $[PdCl_2(Amphos)_2]$ (0.02 mmol), degassed water (2 mL). [b] Determined by GC analysis. [c] 10–15 % of 2-methoxypyridine formed. [d] Yield of isolated product. Bn = benzyl.

[a] V. Krasovskaya, Dr. A. Krasovskiy, Prof. B. H. Lipshutz
Department of Chemistry & Biochemistry
University of California
Santa Barbara, California 93106 (USA)
Fax: (+1) 805-893-8265
E-mail: lipshutz@chem.ucsb.edu

Supporting information for this article is available on the WWW under <http://dx.doi.org/10.1002/asia.201100153>.

Table 2. Representative benzylic-heteroaryl cross-coupling reactions.^[a]

$\text{FG}-\text{C}_6\text{H}_4-\text{CH}_2-\text{R} + \text{Br}-\text{HetAr}-\text{FG}' \xrightarrow[\text{H}_2\text{O, RT, 6 h}]{\text{cat. [PdCl}_2(\text{Amphos})_2], \text{Zn, TMEDA}}$ $\text{FG}-\text{C}_6\text{H}_4-\text{CH}(\text{R})-\text{HetAr}-\text{FG}'$ R = H, Me			
Entry	Product, Yield ^[b]	Entry	Product, Yield ^[b]
1	2 (88 %)	2	3 (94 %)
3	4 (96 %)	4	5 (79 %)
5	6 (81 %) ^[d]	6	7 (90 %)
7	8 (91 %)	8	9 (89 %, 70 %) ^[c]
9	10 (86 %) ^[d]	10	11 (92 %)
11	12 (85 %)	12	13 (87 %)

[a] Conditions: benzylic chloride (3 mmol), heteroaromatic bromide (1 mmol), zinc dust (4 mmol), [PdCl₂(Amphos)₂] (Amphos = *para*-dimethylaminophenyl-di-*tert*-butylphosphine) (0.02 mmol), degassed water (3 mL). [b] Yield of isolated product. [c] From benzylic bromide (3 mmol). [d] Yield determined by NMR analysis.

Table 2 shows a variety of examples based on the conditions developed above. Coupling of both primary 1-(chloromethyl)-4-methylbenzene and secondary (1-chloroethyl)benzene with heteroaromatic bromides afforded the desired products **7** and **8** in high yields within a six hour time frame (Table 2, entries 1–7). Under otherwise-identical conditions, benzylic chlorides afforded higher yields than do benzylic bromides, presumably owing to faster consumption of the bromide and its extensive homocoupling (Table 2, entry 8). As expected, functionalized benzylic chlorides readily participate in the coupling reaction, leading to high yields of the desired arylheteroarylmethanes (Table 2, entries 9–12). Steric effects do not appear to play a major role in these reactions (e.g., Table 2, entry 11).

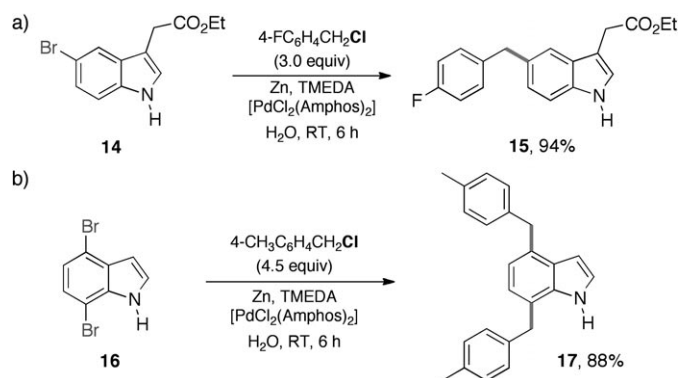
With respect to the nature of the heteroaromatic halide partner, it is worthy of note that a broad spectrum of heterocycles appear to be amenable to this transformation, including the previously^[12] challenging 2-bromothiophenes (Table 2, entries 9 and 12), where very low yields of sp²–sp³ cross-coupled products using alkyl halides owing to competi-

tive zinc insertion were observed. This finding was applied to an efficient synthesis of Plavix derivative **13**. Cross-coupling of 3-bromoquinoline also proceeded cleanly (Table 2, entry 4) without known competitive Chichibabin-like^[13] 1,2-addition of in-situ-formed organozinc reagents to the quinoline core.^[12]

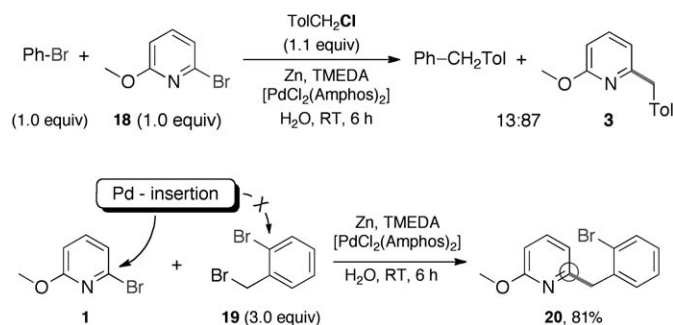
Noteworthy is the cross-coupling of unprotected indole **14**, leading to functionalized indolyl acetic acid derivative **15** in good yield (Scheme 1a). Although under a variety of conditions selective monobenylation of both *N*-Boc (Boc = *tert*-butoxycarbonyl)-protected and -unprotected dibromoin-dole **16** could not be achieved, dibenylation proceeded smoothly leading to difunctionalized indole **17** in 88% yield (Scheme 1b).

Assuming different rates of palladium insertion into heteroaryl versus aryl bromides, competition experiments were performed to test the potential selective benzylation of pyridine **18** in the presence of bromobenzene (Scheme 2). Having observed reasonable selectivity for this unsubstituted case, it was anticipated that *ortho*-substitution on an aryl bromide should further increase the ratio in favor of unsymmetrical arylheteroarylmethane formation. Indeed, cross-coupling of pyridine **18** with three equivalents of commercially available 1-bromo-2-(bromomethyl)benzene (**19**) led to a good yield of isolated 2-(2-bromobenzyl)-6-methoxypyridine (**20**), which contained functionality that can be used in further cross-coupling reactions to build more-complex molecular architectures (Scheme 2).

In conclusion, a new procedure has been developed for preparing functionalized unsymmetrical arylheteroarylmethanes that involves an especially straightforward approach: combining two halide precursors in the presence of inexpensive zinc metal and a palladium catalyst, all in water at room temperature. The couplings take place chemoselectively, to afford the target products in high yields.



Scheme 1. Cross-couplings on unprotected indoles.



Scheme 2. Competition experiment, and chemoselective cross-coupling (Tol = 4-CH₃C₆H₄).

Experimental Section

General Procedure

Degassed water (2 mL) was added to a 5 mL microwave vial containing zinc dust (260 mg, 4 mmol) and [PdCl₂(Amphos)₂] (14 mg, 0.02 mmol) under argon. *N,N,N',N'*-Tetramethylethylenediamine (TMEDA, 116 mg, 1 mmol) was added at RT, followed by the addition of the benzyl halide (3 mmol) and the heteroaryl bromide (1 mmol). The reaction mixture was stirred vigorously at RT for 6 h. The product was extracted with EtOAc. Silica gel (1 g) was added to the combined organic phase and the solvents were removed under vacuum. The resulting dry, crude silica was introduced onto the top of a silica gel chromatography column to purify the product.

Acknowledgements

We are grateful to the NIH for financial support (GM 86485), and to Johnson Matthey for generously supplying palladium catalyst [PdCl₂(Amphos)₂] (Pd-132, catalog #C4138).

Keywords: arylheteroarylmethanes • benzyl halides • cross-coupling • organozinc reagents • water

- [1] a) A. Gangjee, R. Devraj, S. F. Queener, *J. Med. Chem.* **1997**, *40*, 470; b) A. Gangjee, A. Vasudevan, S. F. Queener, *J. Med. Chem.* **1997**, *40*, 3032; c) K. L. McPhail, D. E. A. Rivett, D. E. Lack, M. T. Davies-Coleman, *Tetrahedron* **2000**, *56*, 9391; d) A. Rosowsky, H. Chen, H. Fu, S. F. Queener, *Bioorg. Med. Chem.* **2003**, *11*, 59; e) Y.-

- Q. Long, X.-H. Jiang, R. Dayam, T. Sacher, R. Shoemaker, S. Sei, N. Neamati, *J. Med. Chem.* **2004**, *47*, 2561; f) R. A. Forsch, S. F. Queener, A. Rosowsky, *Bioorg. Med. Chem. Lett.* **2004**, *14*, 1811.
- [2] J. C. Ma, D. A. Dougherty, *Chem. Rev.* **1997**, *97*, 1303, and references therein.
- [3] T. Tsuchimoto, K. Tobita, T. Hiyama, S. Fukuzawa, *J. Org. Chem.* **1997**, *62*, 6997, and references therein.
- [4] a) *Transition Metals in the Synthesis of Complex Organic Molecules*, 3rd ed. (Eds.: L. S. Hege, B. C. G. Söderberg), University Science Books, Sausalito, CA, **2009**; b) *Palladium in Organic Synthesis* (Ed.: J. Tsuji), Springer, Berlin, **2005**; c) E.-I. Negishi, S. Gagneur in *Handbook of Organopalladium Chemistry for Organic Synthesis*, Vol. 1 (Ed.: E. Negishi), Wiley-VCH, New York, **2002**, pp. 597; d) *Metal-Catalyzed Cross-coupling Reactions* (Eds.: F. Diederich, P. G. Stang), Wiley-VCH, New York, **1998**.
- [5] a) E. Negishi, A. O. King, N. Okukado, *J. Org. Chem.* **1977**, *42*, 1821; b) B. Betzemeier, P. Knochel, *Angew. Chem.* **1997**, *109*, 2736; *Angew. Chem. Int. Ed. Engl.* **1997**, *36*, 2623; c) A. Garcia Martinez, J. Osio Barcina, M. del Rosario Colorado Heras, A. de Fresno Cerezo, *Org. Lett.* **2000**, *2*, 1377; d) S.-H. Kim, R. D. Rieke, *J. Org. Chem.* **2000**, *65*, 2322; e) K. Itami, M. Mineno, T. Kamei, J.-I. Yoshida, *Org. Lett.* **2002**, *4*, 3635; f) Y. Suh, J.-S. Lee, S.-H. Kim, R. D. Rieke, *J. Organomet. Chem.* **2003**, *684*, 20; g) K. Park, M. Kang, J. E. Yie, J. M. Kim, L.-M. Lee, *Tetrahedron Lett.* **2005**, *46*, 2849; h) A. Flaherty, A. Trunkfield, W. Barton, *Org. Lett.* **2005**, *7*, 4975; i) M. A. Schade, A. Metzger, S. Hug, P. Knochel, *Chem. Commun.* **2008**, 3046.
- [6] M. Amatore, C. Gosmini, *Chem. Commun.* **2008**, 5019, and references therein.
- [7] a) A. Krasovskiy, C. Duplais, B. H. Lipshutz, *J. Am. Chem. Soc.* **2009**, *131*, 15592; b) A. Krasovskiy, B. H. Lipshutz, *Org. Lett.* **2010**, *12*, 4742; c) W. M. Czaplik, M. Mayer, A. J. von Wangelin, *Angew. Chem.* **2009**, *121*, 616; *Angew. Chem. Int. Ed.* **2009**, *48*, 607.
- [8] C. Duplais, A. Krasovskiy, A. Wattenberg, B. H. Lipshutz, *Chem. Commun.* **2010**, 46, 562.
- [9] a) S. Narayan, M. G. Finn, V. V. Fokin, H. C. Kolb, K. B. Sharpless, *Angew. Chem.* **2005**, *117*, 3339; *Angew. Chem. Int. Ed.* **2005**, *44*, 3275; b) C.-J. Li, *Chem. Rev.* **2005**, *105*, 3095.
- [10] (Amphos)₂PdCl₂ (CAS #887919-35-9) obtained from Johnson Matthey. A. S. Guram, X. Wang, E. E. Bunel, M. M. Faul, R. D. Larsen, M. J. Martinelli, *J. Org. Chem.* **2007**, *72*, 5104.
- [11] a) B. H. Lipshutz, S. Ghorai, *Aldrichimica Acta* **2008**, *41*, 59; b) B. H. Lipshutz, S. Ghorai, A. R. Abela, R. Moser, T. Nishikata, C. Duplais, A. Krasovskiy, R. D. Gaston, R. C. Gadwood, *J. Org. Chem.* in press.
- [12] A. Krasovskiy, I. Thomé, J. Graff, V. Krasovskaya, P. Konopelski, C. Duplais, B. H. Lipshutz, *Tetrahedron Lett.* **2011**, in press.
- [13] A. E. Chichibabin and O. A. Zeide, *Zhur. Russ. Fiz. Khim. Obshch* **1914**, *46*, 1212.

Received: February 14, 2011
Published online: May 30, 2011